

Binary Tongue Coating Classification Report

1. Objective

This study classifies tongue images into two classes, coated and non-coated, using a progressive experimentation pipeline. The purpose is to show how performance changes before and after preprocessing, augmentation, model enhancement, and validation-based post-processing.

All experiments were implemented in Python with a fixed random seed. The same train, validation, and test split was used for every model, and the held-out test set was not used for model selection.

2. Dataset and Reproducibility

The dataset contains 2007 JPG images. Stratified sampling was used to create an 80% training, 10% validation, and 10% test split. The random seed was fixed as 561.

Split	coated	non_coated	Total
train	801	804	1605
validation	100	101	201
test	100	101	201

3. Neural Network and Experimental Pipeline

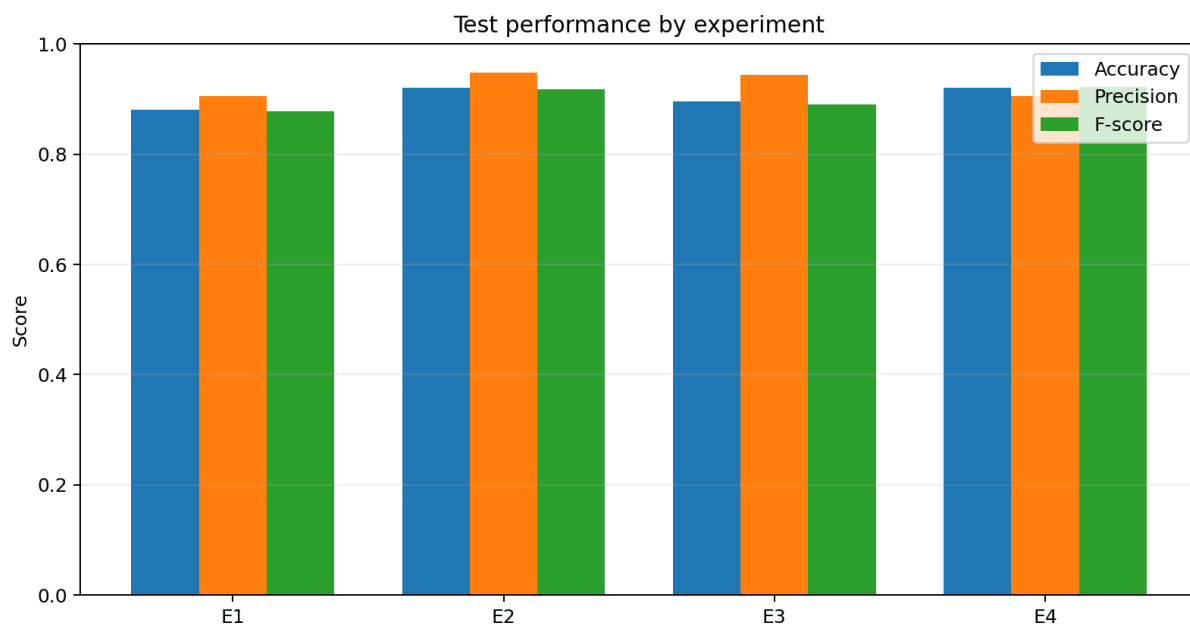
All experiments used an MLP neural network with ReLU activation, Adam optimizer, batch size 32, early stopping, and StandardScaler feature normalization. The internal early-stopping validation fraction was 0.15 of the training split. Augmentation was applied only to the training data.

E1 used resized raw RGB pixels. E2 added gray-world color normalization and CLAHE. E3 added training-only flip, rotation, brightness, and contrast augmentation. E4 used HOG texture features, HSV/color statistics, augmentation, and a validation-selected decision threshold.

Experiment	Hidden	Alpha	LR	Iter	Features	Train	Thr
E1 baseline raw pixels	(64,)	0.0001	0.001	120	12288	1605	0.500
E2 preprocessed pixels	(96,)	0.0001	0.0008	140	12288	1605	0.500
E3 augmented preprocessed pixels	(128, 64)	0.0002	0.0008	160	12288	6420	0.500
E4 hog color postprocessed	(256, 64)	0.0005	0.0005	220	1812	6420	0.200

4. Results

Experiment	Split	Accuracy	Precision	F-score
E1 baseline raw pixels	validation	0.9204	0.9381	0.9192
E1 baseline raw pixels	test	0.8806	0.9053	0.8776
E2 preprocessed pixels	validation	0.8955	0.9000	0.8955
E2 preprocessed pixels	test	0.9204	0.9474	0.9184
E3 augmented preprocessed pixels	validation	0.8955	0.9000	0.8955
E3 augmented preprocessed pixels	test	0.8955	0.9444	0.8901
E4 hog color postprocessed	validation	0.9453	0.9327	0.9463
E4 hog color postprocessed	test	0.9204	0.9048	0.9223



The best test F-score was obtained by E4_hog_color_postprocessed with accuracy 0.9204, precision 0.9048, and F-score 0.9223. Compared with the baseline, the F-score increased by +0.0448.

The baseline model already learned useful visual cues, but preprocessing produced the largest direct gain on the test set. The augmentation experiment did not improve the held-out result, which indicates that the selected transformations were not always beneficial for this dataset. The final HOG/color model improved F-score by changing the balance between precision and recall after threshold selection on the validation set.

5. Conclusion and Future Work

This work followed a systematic before/after pipeline for binary tongue coating classification. Starting with a simple resized-pixel neural network, the pipeline added preprocessing, augmentation, hand-crafted feature enhancement, and validation-only post-processing. The test accuracy increased from 0.8806 in the baseline to 0.9204 after preprocessing. The best final F-score was 0.9223 with the HOG/color feature model and threshold selection.

The results also show why each step should be evaluated separately. Augmentation is usually helpful in image classification, but in this experiment it reduced test performance when applied to pixel features. Therefore, augmentation parameters should be tuned carefully and validated rather than assumed to improve every model.

Future studies can improve accuracy by using a pretrained convolutional neural network with transfer learning, segmenting the tongue region before classification, testing larger hyperparameter grids, using k-fold cross-validation, and applying probability calibration or threshold optimization only on validation data. More controlled image acquisition and a larger labeled dataset would also reduce illumination and color variation, making the learned coating cues more stable.